Production Incentive App — Phase 1 Build Guidelines (Daily Target / Output Batch)

Oct 1, 2026 · @Shahraiz Chishty

1. Context and Phase 1 scope

Give this whole document to Claude as the build brief. Phase 1 is a working front-end prototype, running on dummy data, of the Production / Output KPI (20 of 100 incentive marks) from the DIC Pakistan Limited Production Incentive Policy (revision 30 June 2026). It replaces today's three-sheet Excel process.

What exists today (the Excel the app replaces):

1. *Production Report* sheet: one row per employee, per point, per day. Columns: Date, Emp ID, Name, Point, Daily Target, Output Batch, %age (= output ÷ target), Score (= MIN(%age × 20, 20)), Production Area. About 783 rows, 35 employees, 26 working days (17 Aug to 15 Sep 2026).
2. *Monthly Prod Score Card* sheet: employees down, dates across. Each cell is the average score of that employee on that date. The last column, Total Avg Score, is the average of the daily cells.
3. *Employee KPI* sheet: one row per employee with Production /20, Waste /15, Variance /10, Quality /15, EHS /15, WCM and 5S /15, Attendance /10, Total /100, Approved Incentive and Earned Incentive (= Approved × Total ÷ 100).

In scope for Phase 1

- Daily entry of target and actual batches per employee and point, with automatic score.
- HOD control of daily targets (the policy lets the HOD change the target for order inflow, stock-taking days and machine downtime).
- Monthly scorecard (employee by date grid) and each employee's Production Score out of 20.
- Month-end review, approval and lock, then export for the Finance and Employee KPI sheet.
- Role-based screens, a mock role switcher, an audit trail and realistic dummy data.

Out of scope for Phase 1 (design the data model so they can be added)

- Attendance, Quality, EHS, Variance, Wastage and 5S/WCM marks. Show these only as greyed "Phase 2" placeholders in the employee summary.
- SAP Analyst, Maintenance, Technical, Store, Dispatch and Inplant KPIs (policy sections 6 and 8).
- Real login, a real database, payroll integration and the rupee incentive calculation. A real backend comes after the prototype is tested.

2. Policy rules this phase implements

An employee's daily Production score is their actual batches against target, scaled to 20 marks and capped at 20. The month's Production Score is the average of the daily scores.

$$\text{Daily Score} = \min\left(\frac{\text{Actual Batches}}{\text{Daily Target}} \times 20,\ 20\right)$$

$$\text{Monthly Production Score} = \frac{\sum \text{daily scores}}{\text{number of days with a score}}$$

Rule	Source	How the app applies it
Production / Output is 20% weight, measured as daily actual batches vs target, capped at 20 marks	Policy section 5	Score field is read-only and computed. Never allow a score above 20.
HOD may change the daily target for order inflow, order volume, stock-taking days and machine downtime	Policy section 5 note	HOD-only target edit with a mandatory reason. Every change goes to the audit log.
Applies to Production Operators and Supervisors	Policy sections 2 and 5	Only employees flagged Operator or Supervisor appear in the entry screen.
Office Executive records daily production and compiles the monthly sheet	Policy section 4	Office Executive is the data-entry role.
HOD signs off the monthly sheet. Operations Manager reviews total marking and approves. Finance processes with salary.	Policy section 4	Three-step month-end approval (see section 4 of this brief).

Rule	Source	How the app applies it
Monthly marks must be submitted before the 20th of each month, before the salary run	Policy section 10	Show a countdown banner to the 20th. Warn when the deadline has passed and the month is not submitted.
Adding any employee to the incentive list needs MD approval	Policy section 10	New-employee requests in the app are marked "Pending MD approval" and excluded from the roll-up until approved.
Extraordinary performance: HOD may award full marks (other than the absence deduction)	Policy section 10	HOD "Full marks override" on a day or month, with a reason and an audit entry.
New joiners and leavers are pro-rata (days worked ÷ 26). Probation staff get 50%, with HOD approval. Absence deductions are pro-rata.	Policy section 2.2	Store joining date, leaving date and probation flag on the employee. Show them in the summary but apply them only in the later Employee KPI phase.

Base incentive amounts come from the signed Incentive Revision 2025/2026 sheet (after the increase effective 1 Jan 2025). They are held for later phases and are not used for calculation in Phase 1. Production, grade bands as read from the scan, to be checked against the signed sheet: Non-Kasur (Tetra, Inplant, Karachi) Rs 16,800 / 18,100 / 18,750; Kasur Rs 22,800 / 24,100 / 24,750. Next revision is due Dec 2026 to Jan 2027, so keep base rates in a configurable table, never hard-coded.

3. Roles and permissions

Seven roles, taken from policy section 4 plus one admin role for the app itself. Build a **role switcher** in the header (dummy login) so every role can be tested without real authentication. Every screen and button must respect the matrix below, and a disallowed action must be hidden or disabled with a tooltip explaining why.

Role	Who (policy)	Main job in Phase 1
Office Executive	Records daily production and attendance, compiles the monthly sheet	Enter daily actual batches, submit the month for HOD review

Role	Who (policy)	Main job in Phase 1
HOD (Department Head)	Submits Quality, EHS, WCM marks, reviews and signs off the monthly sheet	Set or change daily targets, review entries, request corrections, sign off the month
Operations Manager	Reviews total marking and approves for further process	Final approval of the month, can reject back to HOD
Finance	Processes the incentive with monthly salary, keeps records	Read-only access to approved months, export
EHS Representative	Provides safety and housekeeping marks	No Phase 1 actions. Read-only view of the scorecard.
MD and CEO	Approves policy revisions, base rate changes, extraordinary awards, additions to the incentive list	Approve new-employee additions and rate or policy changes, read-only everywhere else
System Admin (app role, not in policy)	Maintains master data	Manage employees, points, areas, users and calendar

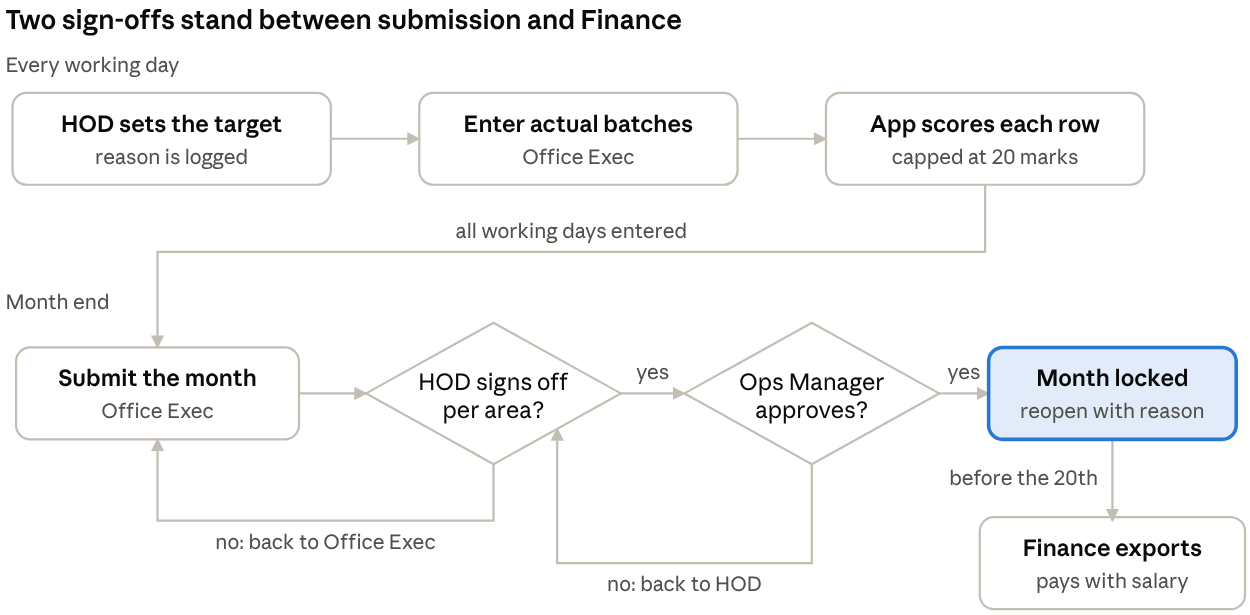
Permission matrix (E = edit, A = approve or sign off, V = view only, none = no access)

Capability	Office Exec	HOD	Ops Manager	Finance	EHS Rep	MD / CEO	Admin
Enter or edit actual batches (open month, unlocked day)	E	E (own area)	none	none	none	none	none
Enter or change daily target	none	E (own area)	none	none	none	none	E (defaults only)
Full marks override with reason	none	E (own area)	none	none	none	A (reviews)	none
View daily entries	V	V (own area)	V	V	V	V	V
View monthly scorecard	V	V	V	V	V	V	V
Submit month to HOD	E	none	none	none	none	none	none
HOD sign-off per area	none	A	none	none	none	none	none

Capability	Office Exec	HOD	Ops Manager	Finance	EHS Rep	MD / CEO	Admin
Operations Manager approval	none	none	A	none	none	none	none
Reopen a locked month	none	none	A (with reason)	none	none	A	none
Export to Finance (CSV or Excel)	V	V	V	E	none	V	V
Manage employees, points, areas, users	none	none	none	none	none	none	E
Approve adding an employee to the incentive list	none	none	none	none	none	A	none
View audit log	none	V (own area)	V	V	none	V	V

HOD scope: a HOD sees and edits only their own Production Area (Solvent, Water Base or Offset). Operations Manager, Finance, MD and Admin see all areas.

4. End-to-end workflows



workflow · daily steps, 2 sign-offs, 2 return loops

Entries are made daily, the month goes up for two sign-offs, and either sign-off can send it back with a comment. Build every step below, and block any attempt to skip a step.

Daily flow (each working day)

1. The HOD reviews the default target for each point. If order inflow, order volume, a stock-taking day or machine downtime changes the day, the HOD edits the target for that date and picks a reason.
2. The Office Executive opens Daily Entry for the date, adds or copies the day's rows, and types the actual batches for each employee and point.
3. The app shows %age and Score live, and runs the validation from section 6 (duplicates, outliers, blank points, zero target).
4. The Office Executive clicks Save draft while working and Submit day when the day is complete. Draft rows can be edited freely. Submitted rows can be changed only by the Office Executive or HOD, and every change writes an audit entry.

Month-end flow

1. When every working day has entries and no warning is open, the Office Executive clicks Submit month. The checklist on the Month-end Review screen must be fully green first.
2. The HOD of each area reviews the area's scorecard and either signs off or returns it with a required comment. A return puts the month back to Open for the Office Executive.
3. When all three areas are signed, the Operations Manager reviews the total marking and either approves or returns it to the HOD with a required comment.
4. On approval the month locks. All entries become read-only and Finance can export.
5. To reopen a locked month, the Operations Manager or MD gives a reason. The month returns to Open and the area HODs are notified.
6. Any new employee added to the incentive list sits as "Pending MD approval" and is excluded from the roll-up until the MD approves, which must happen before the 20th.

Month status and what each allows

Status	Set by	Entries editable by	Next step
Open	Start of period, or any return	Office Exec, HOD (own area)	Office Exec submits the month

Status	Set by	Entries editable by	Next step
Submitted	Office Exec	HOD (own area), only to fix errors	HOD signs off, or returns with comment
HOD Signed (per area)	HOD	Nobody	Ops Manager approves once all areas are signed, or returns to HOD
Locked	Ops Manager approval	Nobody	Finance exports. Reopen only by Ops Manager or MD with a reason.

5. Data model

Keep state in one central store (React context or a small state library) loaded from a seed file, with an optional "Reset demo data" button. Shape the entities as if they will later map to database tables, and use IDs (not names) for every relationship.

Entity	Key fields	Notes
Employee	empId (number, unique), name, grade, site (Kasur / Karachi / Inplant), role (Operator / Supervisor), areaId, status (Active / Probation / Left), joinDate, leaveDate	The Excel stores names inconsistently ("abid", "Abid ", "ABID"). Name is display only. empId is the key.
Area	areaId, name (Solvent, Water Base, Offset), hodUserId	Normalise the Excel spellings ("Solvant", "WATER BASE", "OffSet").
Point	pointId, name, areaId, defaultTarget, active	A point is the work station or activity, for example Dispersion, Grinding, Filtration, Piccolo, Weighing, Varnish Plant, P48. About 30 distinct points exist in the Excel after normalising case.

Entity	Key fields	Notes
DailyEntry	entryId, date, empId, pointId, target, actualBatches, scoreOverride (nullable), overrideReason, enteredBy, enteredAt, updatedBy, updatedAt, status (Draft / Submitted / Locked)	One row per employee, per point, per day. An employee may have several rows in a day. Computed, never stored: percent = actual ÷ target, score = MIN(percent × 20, 20).
TargetRule	ruleId, pointId, effectiveFrom, effectiveTo, target, reason, changedBy	Default target history for a point. A DailyEntry copies the target at creation, so later rule changes never rewrite history.
DayCalendar	date, isWorkingDay, note	Sundays and declared holidays are non-working. 26 is the policy divisor for pro-rata.
MonthPeriod	periodId (YYYY-MM or an incentive cycle), startDate, endDate, status	Status values: Open, Submitted, HOD Signed (per area), Locked (set when the Operations Manager approves). The Excel period runs 17 Aug to 15 Sep, so make start and end dates configurable.
ApprovalStep	periodId, areaId, step (Submit / HOD / Ops), actorUserId, decision, comment, timestamp	One row per action. Supports reject and resubmit.
User	userId, name, role, areaId (nullable)	Dummy users, one or more per role.
AuditLog	logId, timestamp, userId, entity, entityId, field, oldValue, newValue, reason	Written on every create, edit, target change, override, approval and reopen.

Entity	Key fields	Notes
EmployeeMonthSummary (derived)	periodId, empId, daysScored, avgScore, finalScore (0 to 20), flags	Computed from DailyEntry. Not stored by hand.

Derived values the app computes everywhere from the same function (write one pure `calcScore(target, actual)` and one `calcMonthly(entries)` and reuse them in the UI, the export and the tests):

- Daily score per entry, as in section 2.
- Employee-day score = average of that employee's entry scores on that date, as the Excel scorecard does with AVERAGEIFS.
- Monthly Production Score = average of the employee-day scores over days that have at least one entry, rounded to 2 decimals for display and kept unrounded for totals.

6. Calculation rules and edge cases from the real Excel

The current sheet is hand-typed, so it contains errors the app must prevent or flag. Each row below was found in the August to September 2026 file. Build the validation and the demo data around them.

Issue seen in the Excel	Example	Required app behaviour
Same person, many spellings and trailing spaces	"abid", "Abid ", "ABID"; "tahir", " tahir "	Pick the employee from a searchable list by Emp ID. Never free-type a name.
Different people with similar names	Tahir 6905 and Tahir 9360; Shehzad 6289 and Shhzad Mehmood 6290	Always show Emp ID beside the name.
Point names written many ways	"manual", "Manual", "MANUAL"; "P48" and "P48 / Dispensor"; "Grinding" and "Grinding + Loading"	Point is a dropdown from the Point master. Keep an alias map in the seed so the import maps old spellings to one point.
Area spelled inconsistently	"Solvant", "WATER BASE", "Water base", "OffSet"	Area is a fixed dropdown of three values.

Issue seen in the Excel	Example	Required app behaviour
Blank point	21 Aug rows for Jawad Ashraf and Shamshad	Point is required. A blank blocks Save and shows a clear message.
One person on several points in one day	Shehzad on UV Sampling and Filling / Grinding; Abid on Filtration and Dispersion	Allowed. Show all rows for the employee-day grouped together, with the day's average score.
Same person entered twice for the same point and day	Abid appears twice on Dispersion on 17 Aug	Detect and warn "Possible duplicate". Let the user merge or confirm. Do not silently double count.
Wrong Emp ID against the name	19 Aug: ID 8102 (Rafeeq) written as "asghar"	When an ID is picked, show the registered name. Flag a mismatch with the free text if importing.
Output far above target, likely a typo	20 Aug: Umair Yaseen, target 1, output 19 (1,900%)	Soft warning when actual is above 3 times target or target is below 20% of that point's default. Score still caps at 20. Require a confirm click.
Score above 100% is capped	Output 4.75 vs target 4	Show the real percentage (118.8%) but the score as 20. Never 23.75.
Target of zero or blank	Not present in this file, but possible	Block save. Division by zero must never reach the calculation.
Fractional batches	2.33, 2.66, 3.34, 1.33	Allow up to 2 decimals. Reject negatives.
Month is not a calendar month	Data runs 17 Aug to 15 Sep	Period start and end dates are configurable. Do not assume 1st to month-end.

Issue seen in the Excel	Example	Required app behaviour
Sundays and holidays skipped	No rows for 23 Aug	Working days come from DayCalendar. Entry on a non-working day needs a confirm.
Dates stored as text	"17.08.2026"	Store real dates. Display as dd.mm.yyyy to match the office.
Employee with no entries on a day	Absent or on leave	No row, so no score for that day. The Production score is the average over scored days only. Absence is handled by the Attendance KPI in a later phase, and the app must not treat a missing day as zero.
External workbook links in the KPI sheet	VLOOKUP into another file for grade and incentive	Not reproduced. Grade and incentive live in the Employee master.

Open policy questions to show on an "Assumptions" screen and keep visible in the UI. The app must apply the default but make it easy to change in Settings later:

1. When one employee has several rows in a day, the Excel gives each row equal weight (a plain average). The default stays the same. An alternative is weighting by target.
2. The monthly score averages over days scored, not over 26 days. The default stays the same.
3. The score is rounded only for display. Totals use the unrounded value.

7. Screens and UI requirements

Build a responsive single-page app with a left navigation and a top bar. The top bar holds the role switcher, the active period selector, the area filter and a status badge for the period. Office staff mostly use desktop and may use a tablet on the floor, so every screen must work from 768 px wide upward and the entry screen must be fast to use with the keyboard.

#	Screen	Who uses it	What it must do
1	**Dashboard**	All roles	Period status, days entered vs working days, average score by area, count of entries below 70% of target, count of warnings (duplicates, outliers), countdown to the 20th, and a to-do card that changes by role (for example "3 days missing entries" for the Office Executive, "2 areas awaiting your sign-off" for the HOD).
2	**Daily Entry**	Office Exec, HOD	Pick a date and see all employees for the chosen area. Show rows with Emp ID, Name, Point, Target (read-only for the Office Exec), Actual Batches (editable), %age and Score (live, read-only). Add row, delete row (only in Draft), copy previous day's rows as a starting point, keyboard navigation (Tab and Enter move down the Actual column), inline validation per section 6, a day total, and "Save draft" and "Submit day". Locked days are read-only with a lock icon.

#	Screen	Who uses it	What it must do
3	**Targets**	HOD (edit), others view	Table of points with the default target. HOD can set a target for a specific date or date range with a required reason from a list (Order inflow, Order volume, Stock-taking day, Machine downtime, Other). Show the change history per point. A target change never alters locked or already-submitted days without HOD re-opening them.
4	**Monthly Scorecard**	All roles (view)	Reproduce the Excel grid: employees down, dates across, cell = employee-day score, last column = Total Avg Score. Colour scale (red below 12, amber 12 to 17, green above 17, grey for no entry). Sticky first columns and header. Click a cell to open that employee-day's entries. Filter by area and point. Export to CSV.
5	**Employee Detail**	HOD, Ops Manager, Office Exec	Employee card, daily score trend chart for the period, table of every entry, month average, and a Phase 2 placeholder block that shows the other six KPIs as greyed cards with their weights (Quality 15, EHS 15, Wastage 15, 5S and WCM 15, Variance 10, Attendance 10).

#	Screen	Who uses it	What it must do
6	**Month-end Review**	Office Exec (submit), HOD (sign), Ops Manager (approve)	Per area checklist: all working days entered, no open warnings, no draft rows. Buttons depend on role and step. Sign-off and approval require a confirmation dialog and an optional comment. Rejection requires a comment. Show the step history with names and timestamps.
7	**Export**	Finance, Ops Manager	Approved periods only. CSV and XLSX with columns Emp Code, Employee Name, Grade, Area, Days Scored, Production Score /20, laid out so it can be pasted into the existing Employee KPI sheet (column D).
8	**Audit Log**	HOD, Ops Manager, Finance, MD, Admin	Searchable, filterable list: who, when, what changed, old value, new value, reason.
9	**Master Data**	Admin	Employees, points, areas, users and the day calendar, with add, edit and deactivate. Never hard-delete a record that has entries.
10	**Settings and Assumptions**	Admin (edit), all (view)	Score cap (20), pro-rata divisor (26), review deadline day (20), outlier thresholds, period start and end dates, and the open policy questions from section 6.

UI rules that apply everywhere

- Every rule from section 3 is enforced in the UI. A button the role cannot use is hidden or disabled with a tooltip.

- Status badges use text as well as colour: Draft (day rows), Open, Submitted, HOD Signed, Locked.
- Numbers: scores to 2 decimals, percentages to 1 decimal, batches to 2 decimals.
- Empty states, loading skeletons and error messages on every screen.
- Confirm before any destructive or approval action. Toasts confirm success.
- Accessible: labels on inputs, visible focus, sufficient colour contrast, and a keyboard path through the Daily Entry grid.
- Urdu-ready: use a font stack that renders Urdu and keep layouts free of fixed-width text so a later RTL or bilingual version is possible.

8. Dummy data specification

Generate the seed in code (a seeded random generator so results repeat) and keep it in one file. Use **invented names and Emp IDs**, not the real staff from the Excel, but copy the real shape: volumes, areas, point types and the messy patterns in section 6.

Item	Seed size and rule
Period	17 Aug 2026 to 15 Sep 2026 (30 calendar days), Sundays excluded, giving about 26 working days. Today in the app is fixed at 1 Oct 2026 so the 20th deadline is shown as passed for the demo period, and a second open period (16 Sep to 15 Oct) is available for fresh entry.
Areas	Solvent (about 20 employees), Water Base (about 5), Offset (about 10).
Points	About 30, for example Dispersion, Grinding, Grinding + Loading, Filtration, Manual, Varnish Plant, P48 / Dispenser, Piccolo, Pessa + Mixing, Dispersion + Grinding, Weighing, Weighing / Mixing, UV Sampling, UV Filling, Filling / Grinding, UV Dead Stock Charging. Each has a default target between 1 and 15 batches.
Employees	About 35 with 4-digit Emp IDs, grade bands matching the policy (Grade A to P3, Grade 1 to JE-1, TE to TE4), a mix of Kasur, Karachi and Inplant, 2 on Probation, 1 joiner mid-period and 1 leaver.

Item	Seed size and rule
Daily rows	About 780 rows. Most employees have one row a day, about 15% have two rows on different points.
Performance spread	Most days land between 90% and 140% of target, so most scores are 20. About 12% of rows are below target (scores 10 to 18). A few rows are at 50% or less.
Users	At least one per role, one HOD per area (three HODs), two Office Executives, one Operations Manager, one Finance, one EHS Rep, one MD and CEO, one Admin.

Seeded test scenarios (each must be reachable from the UI so a tester can see it without editing code):

1. A normal employee who scores 20 every day.
2. An employee who is under target for a week and averages about 15.
3. A row where output exceeds target (4.75 vs 4) to prove the cap at 20 while still showing 118.8%.
4. A duplicate row for the same employee, point and day.
5. An outlier row (target 1, actual 19) that triggers the confirm warning.
6. A blank-point row that blocks Save.
7. An employee working two points in one day with different scores, to check the day average.
8. A HOD target change on a stock-taking day, with the reason in the audit log.
9. A day missing entries for a whole area, to show the dashboard to-do and the month-end checklist block.
10. A period in each state: Open, Submitted, HOD Signed and Locked, so each role can test its step.
11. A rejected submission with a comment, to test resubmit.
12. A pending request to add a new employee, waiting for MD approval.
13. A probation employee and a mid-period joiner, flagged in the summary for later pro-rata.

9. Technical guidelines, acceptance tests and what comes next

Technical guidelines for Claude

- Build a single-page web app. Preferred stack: React with TypeScript, Tailwind for styling and a small state store. Everything runs in the browser from the seed file, with no backend.
- Keep calculation code in a separate pure module (`calcScore` , `calcEmployeeDay` , `calcMonthly` , `validateEntry`) with unit tests. The UI never re-implements the maths.
- Put policy constants (cap 20, divisor 26, deadline day 20, outlier thresholds) in one config file.
- Write access checks in one `can(role, action, resource)` function and call it from every screen and button. Do not scatter role checks through the components.
- Persist demo changes in memory or browser storage so a tester can refresh. Provide "Reset demo data".
- Every write goes through one function that also writes the audit entry.
- Name files and entities after the policy terms (Production Area, Point, Daily Target, Output Batch) so office staff recognise them.
- Build order: seed data and calculation module, then Daily Entry, Monthly Scorecard, Targets, Month-end Review and Export, then Dashboard, Audit Log, Master Data and Settings.

Acceptance tests (the prototype passes when all are true)

#	Test	Expected result
1	Target 4, actual 4.75	%age 118.8%, score 20.00
2	Target 3, actual 2.5	%age 83.3%, score 16.67
3	Target 5, actual 3	%age 60.0%, score 12.00
4	Target 0 or blank	Save blocked with a message, no NaN or Infinity anywhere
5	Employee with two points in one day, scores 20 and 10	Day score 15.00
6	Employee with day scores 20, 20 and 10	Monthly Production Score 16.67
7	Office Executive tries to edit a target	Control hidden or disabled with a tooltip

#	Test	Expected result
8	Solvent HOD opens the Offset area	Entries not editable, and not visible unless the role allows view
9	Ops Manager approves before HOD has signed	Approve button disabled with the reason
10	Month locked	Entries read-only. Only reopen (Ops Manager or MD, with reason) unlocks them.
11	Any change to actual, target or override	One audit row with old and new value, user and time
12	Scorecard Total Avg Score	Equals the Excel method: average of employee-day scores over scored days
13	Export of an approved period	File matches the on-screen scorecard row for row
14	Duplicate row warning	Appears for same employee, point and day, and blocks submit until confirmed or merged

Planned next phases (do not build now, but do not block them)

1. Attendance marks (days present ÷ 26) and absence pro-rata.
2. Quality, EHS and 5S/WCM marks entered by the HOD and EHS representative, plus the Variance and Wastage tables (policy section 7).
3. Employee KPI total out of 100, Approved Incentive by grade and site, and Earned Incentive = Approved × Total ÷ 100.
4. SAP Analyst and Technical or Support department KPIs (policy sections 6 and 8), where each department submits one final PKR value per employee.
5. Real login, a database, SAP or payroll integration and a mobile-friendly floor entry view.

Questions for DIC to answer before the real build

1. Should a missing day ever count as zero for the Production score, or only through Attendance?
2. Is a plain average across several points correct, or should heavier points count more?
3. Does the HOD target change need Operations Manager approval, or is the HOD's reason enough?

4. Is the incentive period always 17th to 15th, or does it move?
5. Which sites and areas does each HOD cover, and who covers a HOD on leave?
6. The scan marks 50% of incentive on overtime as applicable to Production and Inplant only. Confirm how this interacts with the Production score before Phase 3.